

Revision_pos_neg_generations.ipynb

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Motivation/Background

For performance assessment, positive and negative controls are needed for training and testing models. In the context of the dataset (cancer_complexes) we used, a positive control is defined as the protein-subunit encoding genes that participate in the formation of the same cancer complex, reflecting co-complex membership as a proxy for functional interaction. On the other hand, a negative control is defined as the genes screened by the DepMap project that are not annotated as members of any protein complexes in the Comprehensive Resource of Mammalian Protein Complexes (CORUM).

Important function description in the scripts

PositiveControlGeneration. For each cancer complex in the dataset, there are many subunit-encoding genes that participate. But every complex has distinct participating genes. The genes in each complex were drawn without replacement to ensure no single gene was dominant. Additionally, the upper limit on how many pairs each complex could contribute was set to 5, ensuring that no single cancer complex dominated. The function would randomly shuffle the pairing between genes every time.

NegativeControlGeneration. From all genes that were screened by the DepMap project, a filtered set of 11336 genes was used to generate negative controls. Genes in this filtered set were not annotated to participate in complexes from our cancer dataset or the CORUM dataset.